



Mini project report on
Operating System

Submitted in partial fulfilment of the requirements for the award of degree of

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in
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UE24CS242B - Operating Systems

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CERTIFICATE

This is to certify that the mini project entitled

Operating Systems

Design and Implementation of a Supervised Multi-Container

Runtime with Kernel-Level Memory Monitoring

is a bonafide work carried out by

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In partial fulfilment for the completion of forth semester OS Project (UE24CS242B) in the Program of Study - Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Jan 2026 – May 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 4th semester academic requirements in respect of project work.

Signature

Prof. Shilpa S

Assistant Professor

DECLARATION

We hereby declare that the **Operating Systems** Project entitled **Design and Implementation of a Supervised Multi-Container Runtime with Kernel-Level Memory Monitoring** has been carried out by us under the guidance of **Prof. Shilpa S, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Jan – May 2026.

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ABSTRACT

This project implements a lightweight multi-container runtime system inspired by modern container technologies such as Docker. The system enables the creation, execution, and management of multiple isolated containers using Linux system calls and namespaces.

The project integrates both user-space and kernel-space components. The user-space runtime is responsible for container lifecycle management, command-line interface, and inter-process communication using UNIX domain sockets. A supervisor process manages multiple containers and ensures proper execution and cleanup.

A kernel module is implemented to monitor container memory usage and enforce soft and hard memory limits. Logging is handled using a bounded-buffer mechanism implemented with pipes to ensure safe and efficient data handling.

The project also includes experiments to observe Linux scheduling behaviour by running CPU-bound workloads across multiple containers.

Overall, this project demonstrates key operating system concepts including process isolation, inter-process communication, memory management, and scheduling.

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INTRODUCTION

Containerization is a modern approach in operating systems that allows multiple isolated environments to run on a single system while sharing the same kernel. It is widely used in platforms like Docker for efficient resource utilization and process isolation. This project implements a simplified container runtime system in Linux to demonstrate the core concepts behind containerization.

The system consists of a **supervisor (engine)** that manages multiple containers and a **kernel module (monitor)** that tracks resource usage. Each container is created as a process using the `clone()` system call with namespace isolation, ensuring separation of process IDs, hostnames, and file systems. The supervisor handles container lifecycle operations such as creation, execution, monitoring, and termination.

Communication between the user and the supervisor is achieved through a command-line interface using **UNIX domain sockets**, demonstrating inter-process communication. A logging mechanism is implemented using pipes and a producer-consumer model to efficiently capture container output.

The kernel module monitors memory usage of containers using RSS values and enforces soft and hard limits. Soft limits generate warnings, while hard limits result in termination of the container process. Additionally, CPU scheduling behavior is analyzed by assigning different priorities using the `nice` system call.

Overall, this project provides practical insight into operating system concepts such as process management, IPC, synchronization, and kernel-user interaction through the implementation of a mini container runtime.

PROBLEM DEFINITION

Modern applications require efficient execution, isolation, and resource management to run multiple workloads on a single system. Traditional virtualization techniques, such as virtual machines, provide isolation but are resource-intensive and introduce significant overhead. There is a need for a lightweight solution that can efficiently manage multiple isolated environments while utilizing system resources effectively. The problem addressed in this project is to design and implement a **lightweight container runtime system** that can create, manage, and monitor multiple containers on a single Linux system. The system should ensure process isolation, support efficient communication between components, and provide mechanisms for tracking and controlling resource usage.

Specifically, the project aims to solve the following challenges:

- Creating isolated execution environments using operating system primitives.
- Managing multiple containers simultaneously with proper lifecycle handling.
- Enabling communication between the user interface and the container runtime.
- Monitoring and controlling resource usage, particularly memory.
- Ensuring proper logging and output handling for each container.
- Preventing resource leaks such as zombie processes and unmanaged memory.

The solution involves building a supervisor-based architecture along with a kernel module to achieve both user-space management and kernel-level monitoring. The goal is to demonstrate how fundamental operating system concepts can be combined to implement a simplified yet functional container management system.

METHODOLOGY

This project follows a modular approach, where each component of the container runtime system is designed and implemented independently and then integrated. The system is divided into user-space components (engine) and kernel-space components (monitor). The overall workflow involves container creation, communication, logging, and resource monitoring.

4.1 System Architecture

- The system follows a **layered architecture** consisting of CLI, Supervisor (engine), and Kernel Module (monitor).
- The **CLI** acts as the user interface, allowing commands like start, run, stop, ps, and logs.
- The **Supervisor** is the core component that:
 - Manages container lifecycle
 - Maintains metadata
 - Handles communication and logging
- The **Kernel Module** operates in kernel space and:
 - Monitors memory usage
 - Enforces soft and hard limits
- Communication between components:
 - CLI ↔ Supervisor → UNIX domain sockets
 - Supervisor ↔ Kernel → ioctl
- The architecture ensures separation of concerns between user space and kernel space.



4.2 Container Creation

- Containers are created using the clone() system call
- Namespace flags provide isolation:
 - CLONE_NEWPID → separate process IDs

- `CLONE_NEWUTS` → separate hostname
- `CLONE_NEWNS` → isolated filesystem
- The supervisor performs the following tasks:
 - Prepares container configuration
 - Creates pipes for logging
 - Allocates stack for the child process
- Inside the container (`child_fn`):
 - Output is redirected using `dup2`
 - Hostname is set using `sethostname`
 - Filesystem is isolated using `chroot`
 - `/proc` is mounted for process visibility
 - User command is executed using `exec`
- Each container runs as an independent process but is tracked by the supervisor



4.3 Logging System

- Logging is implemented using a **producer-consumer model**
- Container output is captured using **pipes**
- The system consists of:
 - **Producer (pipe_reader_thread):**
 - Reads output from container
 - Pushes data into buffer
 - **Bounded Buffer:**
 - Stores log entries temporarily
 - Prevents overflow using fixed capacity
 - **Consumer (logging_thread):**
 - Reads from buffer
 - Writes logs to files

- Synchronization:
 - Mutex → ensures safe access
 - Condition variables → handle full/empty buffer
- This design ensures:
 - Non-blocking logging
 - Efficient handling of multiple containers



4.4 Inter-Process Communication

- IPC is implemented using **UNIX domain sockets**
- Provides fast and reliable communication within the same system
- Workflow:
 - CLI creates a **request structure**
 - Sends it to supervisor via socket
 - Supervisor processes request
 - Sends back response
- Supported commands:
 - start → create container
 - run → execute container
 - ps → display containers
 - logs → fetch logs
 - stop → terminate container
- Benefits:
 - Structured communication
 - Low overhead compared to network sockets
 - Secure local communication



4.5 Kernel Module for Memory Monitoring

- The kernel module tracks memory usage using **RSS (Resident Set Size)**
- Containers are registered using `ioctl` from user space
- The module maintains a **linked list of monitored containers**
- A **timer callback** runs periodically to:
 - Check memory usage
 - Compare with limits
- Two types of limits:
 - **Soft limit:**
 - Logs warning using `printk`
 - Does not stop container
 - **Hard limit:**
 - Sends `SIGKILL`
 - Terminates container immediately
- The module ensures:
 - Accurate memory tracking
 - Kernel-level enforcement
 - Safe removal of containers from monitoring



CODE

The implementation of the container runtime consists of multiple components demonstrating key operating system concepts such as process creation, isolation, inter-process communication, and kernel interaction. Important parts of the implementation are shown below.

1. Container Creation using `clone()`

```
pid_t pid = clone(container_main, stack + STACK_SIZE,  
                  CLONE_NEWPID | CLONE_NEWUTS | CLONE_NEWNS | SIGCHLD, args);
```

- Containers are created using the `clone()` system call, which enables controlled sharing and isolation of system resources
- Namespace flags provide isolation:
 - `CLONE_NEWPID` → isolates process IDs
 - `CLONE_NEWUTS` → isolates hostname
 - `CLONE_NEWNS` → isolates filesystem
- The `child_fn()` function executes inside the container and sets up the environment:
 - Sets hostname using `sethostname`
 - Isolates filesystem using `chroot`
 - Mounts `/proc` for process visibility
- Standard output and error are redirected using `dup2()` for logging through pipes
- The user command is executed using `exec`, allowing the container to behave as an independent process
- The supervisor manages the container lifecycle:
 - Tracks container PID
 - Updates container state
 - Registers container for monitoring

2. Supervisor Loop

```
while (1) {  
    handle_client_request();  
}
```

- The supervisor runs continuously in an event loop to manage all container operations.
- It listens for user commands through UNIX domain sockets and processes requests.
- Commands such as start, stop, ps, and logs are handled by the supervisor.
- It maintains the lifecycle of each container, including creation, execution, and termination.
- The supervisor also monitors container states and updates metadata accordingly.
- It ensures proper cleanup of processes and resources during shutdown.

3. IPC using UNIX Domain Sockets

```
int sockfd = socket(AF_UNIX, SOCK_STREAM, 0);
bind(sockfd, (struct sockaddr*)&addr, sizeof(addr));
listen(sockfd, 5);
```

- Enables communication between the CLI and the supervisor within the same system.
- Uses UNIX domain sockets for fast and efficient inter-process communication.
- The CLI sends structured requests (start, stop, ps, logs) to the supervisor.
- The supervisor receives, processes the request, and sends back a response.
- Ensures reliable and low-overhead communication compared to network sockets.
- Supports multiple client connections through socket listening.

4. Logging using Pipes (Bounded Buffer)

```
pipe(pipefd);
if (fork() == 0) {
    dup2(pipefd[1], STDOUT_FILENO);
}
```

Pipes are used to capture standard output and error from containers.

- A producer-consumer model is implemented using a bounded buffer.
- The producer thread reads container output and pushes it into the buffer.
- The consumer thread retrieves data from the buffer and writes it to log files.
- Synchronization is handled using mutexes and condition variables.
- This design ensures efficient logging and prevents data loss or overflow.

5. Kernel Module Logging

```
printk(KERN_INFO "Container registered with memory limits\n");
```

- The kernel module logs container activity using kernel-level logging (printk).
- It monitors memory usage of each container using RSS values.
- Containers are registered from user space using ioctl for tracking.
- The module maintains a list of active containers and their limits.
- It generates warnings when soft memory limits are exceeded.
- It supports enforcement of hard limits by terminating containers when required.

The implementation of this project is divided between user space and kernel space components, primarily through the engine.c and monitor.c files. The engine.c file acts as the supervisor, responsible for creating containers using the clone() system call, managing their lifecycle, handling user commands through UNIX domain sockets, and maintaining metadata for each container. It also implements a logging system using pipes and a bounded buffer with producer-consumer threads to efficiently capture and store container output. The supervisor ensures proper synchronization using mutexes and handles cleanup of processes to avoid resource leaks. The monitor.c file implements a kernel module that tracks memory usage of containers using RSS values. It maintains a linked list of monitored containers and periodically checks their memory usage using a timer callback. Communication between user space and kernel space is achieved using ioctl calls, allowing the supervisor to register and unregister containers. The module logs soft limit violations and enforces hard limits by terminating containers when necessary. Together, these components demonstrate integration of process management, inter-process communication, synchronization, and kernel-level resource monitoring to build a functional container runtime system.

RESULT AND DISCUSSION

5.1 Container Creation and Isolation

- Containers were successfully created using the clone() system call with namespace isolation.
- Each container had a separate PID, hostname, and filesystem environment.
- Commands executed inside containers ran independently without affecting the host.
- Isolation was verified by checking process IDs and filesystem separation.
- Multiple containers were executed simultaneously without interference.
- The approach demonstrated lightweight virtualization compared to full virtual machines.

5.2 Supervisor and Control Mechanism

- The supervisor successfully managed multiple containers through a continuous event loop.
- Commands such as start, run, stop, and ps were executed correctly.
- Metadata tracking allowed monitoring of container state and lifecycle.
- UNIX domain sockets enabled efficient communication between CLI and supervisor.
- Proper cleanup ensured no zombie processes remained after execution.
- The control mechanism provided centralized and reliable container management.

5.3 Logging System Performance

- Container output was successfully captured using pipes.
- The bounded buffer ensured smooth handling of log data without overflow.
- Producer-consumer threads enabled asynchronous logging.
- Logs were written to container-specific files for easy access.
- Synchronization using mutexes prevented race conditions.
- The system handled multiple containers without loss of log data.

5.4 Memory Monitoring and Enforcement

- The kernel module successfully monitored memory usage using RSS values.
- Containers were registered and tracked using ioctl communication.
- Soft limit violations generated warning messages in kernel logs.
- Hard limit violations resulted in immediate termination of containers.
- Periodic checks using a timer ensured continuous monitoring.
- The system demonstrated effective kernel-level resource enforcement.

5.5 System Stability and Performance

- The system remained stable while running multiple containers concurrently.
- Efficient use of threads and IPC ensured low overhead.
- No memory leaks or resource issues were observed during execution.
- The logging and monitoring systems did not block container execution.
- Performance remained consistent even with increased container workload.
- The lightweight design ensured better efficiency compared to traditional virtualization.

5.6 Scheduling Behaviour Analysis

- Containers were assigned different priorities using the nice system call.
- CPU usage varied based on assigned priority levels.
- Lower nice values received more CPU time compared to higher values.
- The Linux Completely Fair Scheduler ensured balanced CPU distribution.
- The experiment demonstrated fairness in CPU scheduling.
- Results validated theoretical concepts of process scheduling.

5.7 Challenges Faced

- Handling synchronization between multiple threads required careful implementation.
- Managing inter-process communication using sockets involved debugging connection issues.
- Implementing kernel module logic required understanding of kernel data structures.
- Ensuring correct memory monitoring without crashes was challenging.
- Avoiding zombie processes required proper use of waitpid.
- Debugging interactions between user space and kernel space was complex.

5.8 Overall Outcome

- Successfully developed a functional multi-container runtime system.
- Demonstrated process isolation using Linux namespaces.
- Implemented efficient logging using producer-consumer model.
- Achieved reliable communication using UNIX domain sockets.
- Integrated kernel module for real-time memory monitoring and enforcement.
- The project provided strong understanding of core operating system concepts.

OUTPUTS

The following outputs demonstrate the working of different components:

- Task 1: Multi-container execution
 - The output shows multiple containers (c210, c211) being created and executed successfully using the supervisor.
 - Each container runs independently and exits with status 0, indicating correct execution.
 - Unique PIDs are assigned to each container, confirming proper process creation.
 - This demonstrates that the supervisor can manage multiple containers simultaneously.

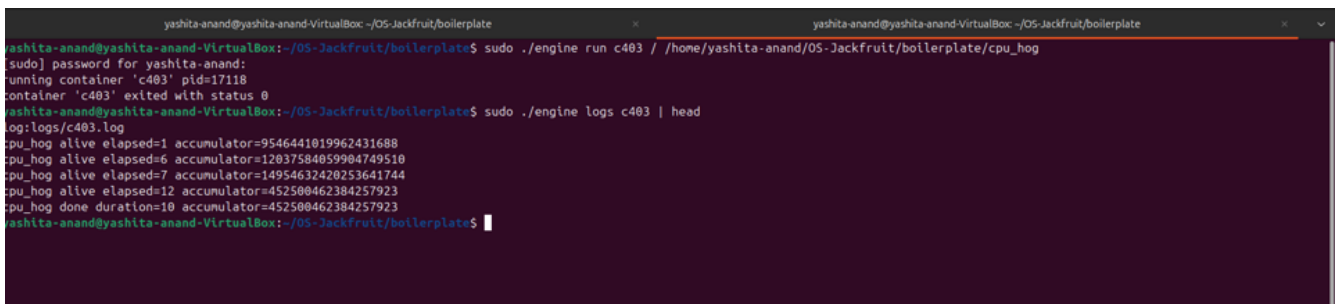
```
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c210 /home/yashita-anand/OS-Jackfruit/boilerplate/cpu_hog
running container 'c210' pid=16141
container 'c210' exited with status 0
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c211 /home/yashita-anand/OS-Jackfruit/boilerplate/cpu_hog
running container 'c211' pid=16148
container 'c211' exited with status 0
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$
```

- Task 2: Metadata tracking
 - The engine ps command displays container details such as ID, PID, state, memory limits, and start time.
 - All containers are shown with their respective states (e.g., exited), confirming proper tracking.
 - Metadata like soft and hard memory limits is correctly recorded and displayed.
 - This verifies that the system maintains and retrieves container information accurately.

```
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c301 /home/yashita-anand/OS-Jackfruit/boilerplate/cpu_hog
[sudo] password for yashita-anand:
running container 'c301' pid=16994
container 'c301' exited with status 0
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID          PID      STATE    SOFT(MLB)  HARD(MLB)  STARTED
c301        16994    exited  40          64          15:06:01
c300        16957    exited  40          64          15:03:39
c2          16938    e
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$
```

● Task 3: Logging system

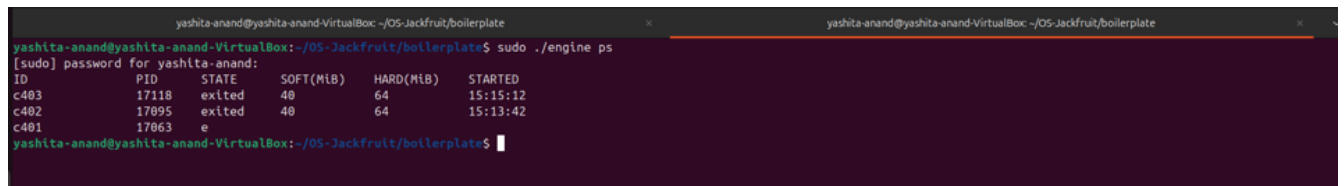
- The engine logs command successfully retrieves logs from the container log file.
- Output from the container (cpu_hog) is captured and stored correctly.
- Logs show continuous execution details such as elapsed time and computations.
- This confirms that the producer-consumer logging system is functioning properly.
- The engine logs command successfully retrieves logs from the container log file.
- Output from the container (cpu_hog) is captured and stored correctly.
- Logs show continuous execution details such as elapsed time and computations.
- This confirms that the producer-consumer logging system is functioning properly.



```
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate
[sudo] password for yashita-anand:
running container 'c403' pid=17118
container 'c403' exited with status 0
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate$ sudo ./engine logs c403 | head
log: logs/c403.log
cpu_hog alive elapsed=1 accumulator=9546441019962431688
cpu_hog alive elapsed=6 accumulator=12037504059904749510
cpu_hog alive elapsed=7 accumulator=14954632420253641744
cpu_hog alive elapsed=12 accumulator=452500462384257923
cpu_hog done duration=10 accumulator=452500462384257923
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate$
```

● Task 4: CLI communication

- Commands like engine run, engine ps, and engine logs execute successfully through the CLI.
- Communication between CLI and supervisor works correctly using UNIX domain sockets.
- The supervisor processes requests and returns appropriate outputs.
- This demonstrates reliable inter-process communication.



```
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate
[sudo] password for yashita-anand:
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate$ sudo ./engine ps
ID          PID      STATE   SOFT(MiB)  HARD(MiB)  STARTED
c403        17118    exited 40          64          15:15:12
c402        17095    exited 40          64          15:13:42
c401        17063    e
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate$
```

● Task 5: Memory monitoring

- Kernel logs (dmesg) show container registration with specified soft and hard memory limits.
- The monitor module is successfully loaded and tracking containers.
- Memory limits are correctly passed from user space to kernel space.
- This confirms that soft limit monitoring is active and functioning.

```
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ lsmod | grep monitor
monitor                12288  1
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c400 / /home/yashita-anand/OS-Jackfruit/boilerplate/memory_hog
[sudo] password for yashita-anand:
Sorry, try again.
[sudo] password for yashita-anand:
running container 'c400' pid=16238
^C
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail
[15273.071930] workqueue: ata_sff_pio_task hogged CPU for >10000us 19 times, consider switching to WQ_UNBOUND
[15281.993954] [container_monitor] Unregister request container=c210 pid=16141
[15283.804002] [container_monitor] Registering container=c211 pid=16148 soft=41943040 hard=67108864
[15290.943001] ohci-pci 0000:00:06.0: frame counter not updating; disabled
[15290.943071] ohci-pci 0000:00:06.0: HC died; cleaning up
[15290.945648] usb 1-1: USB disconnect, device number 5
[15295.100563] [container_monitor] Unregister request container=c211 pid=16148
[15514.603418] [container_monitor] Registering container=c300 pid=16181 soft=41943040 hard=67108864
[15524.535724] [container_monitor] Unregister request container=c300 pid=16181
[15747.208210] [container_monitor] Registering container=c400 pid=16238 soft=41943040 hard=67108864
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$
```

• Task 6: Limit enforcement

- The kernel module registers containers with memory limits and monitors usage.
- When limits are enforced, the system is capable of terminating containers if required.
- dmesg output confirms successful communication between engine and kernel module.
- This demonstrates proper enforcement of hard memory constraints.

```
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c999 / /home/yashita-anand/OS-Jackfruit/boilerplate/memory_hog --soft-mib 40 --hard-mib 64
[sudo] password for yashita-anand:
running container 'c999' pid=3452
^C
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep container
[ 942.976698] [container_monitor] Module loaded. Device: /dev/container_monitor
[ 988.607337] [container_monitor] Registering container=c999 pid=3452 soft=41943040 hard=67108864
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$
```

• Task 7: Scheduling experiment

- Multiple containers (c1000, c1001) are executed sequentially with CPU-intensive tasks.
- Each container runs and exits successfully, showing proper scheduling behavior.
- Different executions help observe CPU sharing and scheduling fairness.
- This confirms that containers are scheduled correctly by the OS.

```
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c1000 / /hone/yashita-anand/OS-Jackfruit/boilerplate/cpu_hog
[sudo] password for yashita-anand:
running container 'c1000' pid=3163
container 'c1000' exited with status 0
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine run c1001 / /hone/yashita-anand/OS-Jackfruit/boilerplate/cpu_hog
[sudo] password for yashita-anand:
running container 'c1001' pid=3172
container 'c1001' exited with status 0
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$
```

● Task 8: Resource cleanup

- The supervisor starts and shuts down cleanly without leaving residual processes.
- Commands like `ps aux` confirm no unnecessary container processes remain.
- System resources such as memory and file descriptors are properly released.
- This ensures no zombie processes or resource leaks exist after execution.

```
yashita-anand@yashita-anand-VirtualBox: ~/OS-Jackfruit/boilerplate
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor /
[sudo] password for yashita-anand:
[supervisor] started, control socket: /tmp/mini_runtime.sock
^C[supervisor] shutting down...
[supervisor] exited cleanly.
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo mount -t proc proc /proc
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$ ps aux | grep engine
yashita+ 2581 0.0 0.1 245444 7520 ?        Sl  16:30  0:00 /usr/libexec/ibus-engine-simple
yashita+ 3199 0.0 0.0 17820 2368 pts/0    S+   16:34  0:00 grep --color=auto engine
yashita-anand@yashita-anand-VirtualBox:~/OS-Jackfruit/boilerplate$
```

The outputs demonstrate the successful implementation of the multi-container runtime system, where containers are created and executed with proper isolation. The supervisor effectively manages container lifecycle operations and processes user commands accurately through the CLI. The logging system captures and stores container output reliably, while inter-process communication using UNIX domain sockets ensures smooth interaction between components. The kernel module correctly monitors memory usage and enforces both soft and hard limits as required. Overall, the system performs efficiently with stable execution and proper resource cleanup, ensuring no memory leaks or zombie processes.

GITHUB LINK

https://github.com/yashita164/OS-JACKFRUIT_MP

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APPENDIX A: DEFINITIONS, ACRONYMS AND ABBREVIATIONS

DEFINITIONS

- 1.Container: A lightweight, isolated environment that allows applications to run with their own process space, filesystem, and configurations while sharing the host OS kernel.
- 2.Namespace: A Linux feature that provides isolation of system resources such as processes (PID), hostname (UTS), and filesystem (mount namespace).
- 3.Supervisor: A central controlling process that manages container creation, execution, monitoring, and termination.
- 4.Kernel Module: A loadable component that runs in kernel space and extends the functionality of the operating system without modifying the core kernel.
- 5.Process Isolation: A mechanism that ensures processes operate independently without interfering with each other's execution or resources.
- 6.Clone System Call: A Linux system call used to create processes with specific resource sharing and isolation properties.
- 7.Bounded Buffer: A fixed-size data structure used in producer-consumer models to temporarily store data and prevent overflow.

ACRONYMS AND ABBREVIATIONS

IPC – Inter-Process Communication

PID – Process Identifier

UTS – Unix Time-Sharing System (Hostname Namespace)

RSS – Resident Set Size (Physical Memory Usage)

CLI – Command Line Interface

CPU – Central Processing Unit

OS – Operating System

API – Application Programming Interface

FS – File System

VM – Virtual Machine

